

IT3051 | FUNDAMENTALS OF DATA MINING | INDIVIDUAL VIVA

Progress Evaluation 1

Four-member preparation guide: problem understanding, dataset validation, EDA, preprocessing, and feature engineering

30%

individual assessment weight

Stages 1-4

required project coverage

**4 equal
members**

everyone explains the full work

Project: Obesity Risk Intelligence System | **Branches:** member-1, member-2, member-3, member-4
Educational classification system only. It is not a medical diagnosis tool.

1. Evaluation 1 Overview

Assessment format

Weight: 30%

Type: Individual viva during the laboratory session

Scope: Everything completed up to preprocessing

Guideline dates

23, 26, or 27 September 2026

KandyUni and Northern Uni dates will be communicated by staff.

The evaluator may ask any member about the group's work, not only the files committed by that member. Each student must demonstrate independent understanding and explain their own technical decisions and contribution.

Scope boundary: For Evaluation 1, do not claim model-training results, hyperparameter-tuning results, backend functionality, or frontend functionality as completed evidence. Those belong to later stages.

What the viva assesses

- Problem/scenario understanding
- Dataset selection and justification
- Dataset characteristics and target variable
- EDA findings and data-quality issues
- Preprocessing techniques and reasons
- Feature engineering or selection decisions
- Data leakage risks and prevention
- Ability to explain personal contribution and the complete group workflow

2. Project Story Every Member Must Know

We are developing an educational multi-class classification system that predicts an obesity-risk category from demographic, physical, nutrition, and lifestyle variables. The target is NObeyesdad. The system supports data-mining learning and decision discussion; it does not diagnose obesity or recommend treatment.

Question	Expected project answer
What is the problem?	Classify a record into an obesity-risk category using available demographic, physical, eating, and lifestyle features.
Who benefits?	Students, researchers, educators, and demonstration users exploring an explainable data-mining workflow.
What is the task?	Supervised multi-class classification.
What is the target?	NObeyesdad, a categorical obesity-risk label.
What is the output?	A predicted category and, later, supporting confidence/probability information with an educational disclaimer.

3. Stage 1: Problem Understanding

Required explanation

- State the real-world scenario in one clear sentence.
- Define the prediction objective and target variable.
- Identify users, stakeholders, decisions, inputs, and outputs.
- Explain why this is classification rather than regression.
- State the ethical and non-clinical limitations.

Good viva answer structure: Problem → user → input → target/output → task type → decision support → limitation.

Evidence to bring

- Problem statement in the project documentation
- Stakeholder and user-requirements table
- Input/output description
- Short ethical and scope statement

4. Stage 2: Dataset Selection and Validation

Dataset proposal

Item	Prepared answer
Name/source	Kaggle Playground Series Season 4 Episode 2 obesity-risk dataset
URL	Kaggle competition data page
Expected size	20,758 training records and 18 columns; verify using the downloaded approved file
Target	NObeyesdad
Task	Seven-class classification; verify the observed class labels in the source file
Feature groups	Physical/demographic, nutrition, hydration, activity, technology use, and transport/lifestyle variables
Why suitable?	Clear target, varied predictive features, enough records for EDA and preprocessing, and suitable complexity for later algorithm comparison

Approval rule: Instructor validation must be recorded before the team proceeds with EDA, preprocessing, model training, or system development. Expected dimensions are not verified findings until checked from the approved download.

Limitations to explain

- Public competition data may be synthetic or simulated.
- Dataset labels are not clinical diagnoses.
- Results may not generalize to every population.
- Do not include Kaggle credentials or private data in the repository.
- Confirm the live page's citation and license details before submission.

5. Stage 3: Data Understanding and EDA

EDA findings must be based on the approved downloaded file. The following are required checks, not invented results. Replace every "to verify" item with measured evidence, tables, and plots.

Investigation	What to check	Evidence
Structure	Rows, columns, names, data types, memory use	Data summary and schema table
Missing data	Null counts and missingness patterns per column	Missing-value table and observation
Duplicates	Duplicate rows and duplicate identifiers	Counts and handling decision
Target	Observed classes, counts, percentages, imbalance	Frequency table and class plot
Numerical variables	Range, quartiles, spread, unusual values, distributions	Summary statistics and plots
Categorical variables	Unique values, spelling/whitespace issues, rare levels	Frequency tables and cleaning notes
Relationships	Feature-target and feature-feature patterns	Selected plots with interpretation
Validity	Impossible ages, heights, weights, or out-of-range survey values	Rule-based validation counts
Leakage	Identifiers, target-derived variables, or test information entering training	Leakage risk register

How to explain an EDA finding

1. State the measured result.
2. Explain why it matters for modelling.
3. State the preprocessing or modelling decision it caused.
4. State how the decision will be validated.

Do not say: "There are no missing values" unless the notebook output proves it. Say: "The approved file was checked, and the missing-value table showed [verified result]. Therefore we [decision]."

6. Stage 4: Preprocessing and Feature Engineering

Decision area	What each member must justify
Identifier	Why id is excluded or retained; default plan is exclusion because it identifies records.
Missing values	Why rows are removed or values are imputed, based on amount and meaning of missingness.
Duplicates	Whether duplicates are removed and how this affects target distribution.
Invalid values	Which domain/range rules are used and why invalid records are corrected, removed, or investigated.
Outliers	Whether an unusual value is an error or a legitimate observation; avoid automatic deletion.
Categorical encoding	Which encoding is suitable and how unseen categories will be handled.
Numerical scaling	Whether the selected algorithms require scaling and where scaling is fitted.
Feature engineering	Which derived features are proposed, why they may help, and how leakage is avoided.
Split strategy	Use a training/test split appropriate for classification, preferably stratified; fit transformations only on training data.

7. Equal Four-Member Division

Each member has a lead area, but all four members share responsibility for the complete Evaluation 1 scope. Every member must review the other members' work and be able to explain it.

MEMBER 1 member-1

Lead: Problem and dataset proposal

Stage 1 Stage 2

- Draft problem, users, stakeholders, target, task type, inputs, and outputs.
- Prepare dataset source, citation, suitability, limitations, ethics, and validation proposal.
- Review EDA and preprocessing evidence for alignment with the scenario.
- Contribute to the complete viva answers and personal contribution record.

Must explain: why this problem and dataset are appropriate, and why the target is classification.

MEMBER 2 member-2

Lead: Data understanding and EDA

Stage 3

- Prepare verified schema, data types, missing/duplicate checks, target distribution, and visualisations after approval.
- Document data-quality observations, invalid values, outliers, relationships, and class balance.
- Review the dataset proposal and preprocessing decisions.
- Contribute to the complete viva answers and personal contribution record.

Must explain: what was found in the data and how each finding influences later preprocessing.

MEMBER 3 member-3

Lead: Preprocessing and feature engineering

Stage 4

- Prepare the evidence-based cleaning, encoding, scaling, split, leakage, and feature-engineering plan.
- Ensure preprocessing is fitted only on training data and is reproducible.
- Review EDA findings and confirm each transformation has a reason.
- Contribute to the complete viva answers and personal contribution record.

Must explain: what each preprocessing technique does, why it is needed, and how leakage is prevented.

MEMBER 4 member-4

Lead: Integration and Evaluation 1 evidence

Cross-stage

- Maintain the requirements checklist, validation evidence, decision log, branch reviews, and contribution register.
- Check that EDA and preprocessing outputs are consistent and reproducible.
- Coordinate the group viva rehearsal; do not replace technical contributions.
- Contribute to the complete viva answers and personal contribution record.

Must explain: how the evidence maps to the assignment requirements and how the team prevents unsupported claims.

Equal assessment: A member cannot answer only their assigned section. The evaluator may ask Member 1 about preprocessing, Member 2 about the dataset proposal, Member 3 about EDA, or Member 4 about any technical decision.

8. Viva Questions and Answer Points

Use these questions for individual rehearsal. Answer in your own words, using verified project evidence rather than memorised general definitions.

1. What problem does your system solve?

State the obesity-risk category classification scenario, intended users, input information, target output, and educational limitation.

2. Why is this classification and not regression?

The target is a set of named categories in `NObeyesdad`, not a continuous numerical quantity.

3. Why did you select this dataset?

Explain target clarity, relevant feature groups, sufficient records, public accessibility, task complexity, and suitability for the proposed scenario.

4. What is your target variable?

`NObeyesdad`; explain its meaning and list the observed classes only after verifying the approved file.

5. How many rows and columns are present?

Give the measured result from the downloaded approved file. The proposal expectation is 20,758 rows and 18 columns, but this must be verified.

6. Which features are numerical and categorical?

Identify the groups from the verified schema and explain why their treatment differs.

7. What did your EDA reveal?

Give two or three measured findings, their supporting plots/tables, and the decisions they caused. Do not invent findings.

8. Did you find missing values or duplicate records?

Cite the actual counts and explain the selected handling policy.

9. How did you investigate outliers and invalid values?

Describe range checks, plots or summary statistics, and why an observation is treated as an error or legitimate value.

10. Is the target balanced?

Use verified class counts/percentages and explain whether stratification, class weights, or macro-averaged metrics may be needed later.

11. Why should `id` normally be excluded?

It is an identifier, not a meaningful subject characteristic, and can create spurious patterns or leakage.

12. What preprocessing will you apply?

Name only decisions supported by the verified data: missing-value handling, duplicate/invalid handling, encoding, scaling, feature engineering, and split strategy.

13. What is data leakage?

Information unavailable at prediction time, or information from validation/test data, incorrectly influencing training or preprocessing.

14. How will you prevent leakage?

Separate data before fitting transformations, fit preprocessing on training data only, exclude target-derived features, and keep the test set untouched until evaluation.

15. Why use a stratified split?

To preserve class proportions across training and test sets for a multi-class target, when the data and split strategy support it.

16. What is feature engineering?

Creating or transforming variables to represent useful information more effectively; explain the proposed feature, reason, and leakage check.

17. What was your personal contribution?

Name exact files, decisions, commits, reviews, evidence, and what you learned. Do not describe only the group's work.

18. What would you improve next?

Mention later model comparison, tuning, error analysis, backend/frontend integration, testing, and stronger validation, without claiming they are already complete.

9. Final Preparation Checklist

Seven days before viva

- ☐ Read the approved dataset proposal and assignment requirements.
- ☐ Run or inspect the data-understanding outputs and replace all pending values.
- ☐ Review every preprocessing decision and its evidence.
- ☐ Read all other members' work, not only your own branch.
- ☐ Prepare a 60-second explanation of the problem and a 60-second explanation of your contribution.

Before entering the viva

- ☐ Know the dataset source, target, feature groups, row count, column count, and verified quality results.
- ☐ Know the reason for every cleaning, encoding, scaling, splitting, and feature decision.
- ☐ Know at least two leakage risks and how the team prevents them.
- ☐ Know the exact files and evidence you personally contributed.
- ☐ Be honest about work that is planned rather than completed.

Answer method

Use evidence

"We checked X and observed Y. Because of that, we chose Z."

Show understanding

Explain the reason, trade-off, limitation, and validation method in your own words.

10. Required Evidence Folder Plan

Evidence	Suggested location
Problem and dataset proposal	docs/
Instructor validation record	docs/validation/ or approved evidence location
Data understanding notebook	notebooks/01_data_understanding.ipynb
EDA figures and summaries	reports/eda/
Preprocessing plan and decisions	docs/ and later src/
Personal contribution logs	docs/contributions/
Branch reviews and decisions	Pull requests and project decision log

Final reminder: Evaluation 1 covers the work up to preprocessing. Every member is equally responsible for understanding and defending the complete work. Verify all dataset-specific numbers from the approved source before the viva.